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# AI in teacher education: Unlocking new dimensions in teaching support, inclusive learning, and digital literacy

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## Abstract

**Background:** AI can positively influence teaching by offering support for classroom management, creating inclusive learning environments, enhancing digital skills, personalizing teaching methods, and strengthening teacher-student relationships.

**Objectives:** This quantitative research study investigates the opportunities, difficulties, and consequences of incorporating AI into teacher education.

**Methods:** Data were collected through structured questionnaires from 202 college students and 68 staff members. The analysis was conducted using SPSS software.

**Results:** The study provides a novel contribution by its thorough investigation of the diverse effects of AI on teacher education. It offers beneficial perspectives on the possible benefits and challenges, illuminating the far-reaching changes that AI could bring to the terrain of learning and instruction and teaching methods in the time yet to come. The research sought to assess the effect of AI adoption in teacher education across five main dimensions: (i) its influence on teaching support and classroom management, (ii) its role in creating inclusive and accessible learning environments, (iii) its contribution to improving teachers' digital literacy and computer skills, and enhancing access to digital teaching resources, (iv) its positive influence on identifying students' learning styles and facilitating the adoption of diverse teaching methods, and (v) its role in strengthening teacher-student relationships through improved interactions.

**Conclusion:** The findings elucidate the promising opportunities that AI presents in the field of teacher education, along with the obstacles that require resolution for the effective fusion of AI educational settings.

KEYWORDS
artificial intelligence, digital literacy, inclusive learning, personalized teaching, student-teacher interaction, teacher education

## 1 | INTRODUCTION

Artificial Intelligence (AI) has surfaced as a revolutionary influence throughout multiple industries, including the realm of education. AI-driven tools possess the capability to enable personalized education experiences for aspiring teachers, helping them acquire the necessary knowledge and skills at their own pace. These tools can also streamline administrative tasks, allowing instructors to concentrate on more meaningful facets of instruction and mentorship (van der Vorst & Jelicic, 2019). AI can enhance professional development through offering real-time feedback and data-driven insights, which can empower educators to continually improve their pedagogical practices. Nonetheless, the incorporation of AI into teacher learning imparts hurdles, with a key issue revolving around the imperative for

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robust data privacy and security measures. This is particularly significant given that the gathering and examination of sensitive educational data are integral to the functionality of AI. Moreover, there is a risk associated with depending too much on technology, potentially reducing the significance of human interaction and mentorship in teacher training. The digital divide, and unequal access to technology, can also make worse educational disparities (Akgun & Greenhow, 2022). AI holds the potential to transform the way educators are prepared, to effectively address the evolving needs of 21st-century learners. However, it is essential to strike an equilibrium balancing technology with human interaction, highlighting that AI ought to serve as an instrument to aid and enhance instead of supplanting teachers. Furthermore, the ethical and societal ramifications of AI, such as concerns related to algorithms and job displacement concerns, must be carefully addressed (Schiff, 2022). AI offers promising opportunities for teacher education; nevertheless, it poses a mass of challenges and implications that demand careful deliberation. Celik et al. (2022) undertook a comprehensive analysis to scrutinize the potential and obstacles of AI in teaching practice from teachers' perspectives and roles in AI-based education research. The findings underscored AI's potential for enhancing teaching planning, implementation, and assessment, while also emphasizing challenges related to technical capacity, reliability, and the necessity for teacher involvement in AI development. Salas-Pilco et al. (2022) carried out a methodical survey of literature encapsulating recent advancements in AI and LA (Learning Analytics) research in teacher education, revealing a predominant focus on teachers in training and those already in service behaviours, perceptions, and technological proficiency, alongside the utilization of machine learning algorithms and moral deliberations. Ng et al. (2023) conducted a conceptual analysis to align AI competencies with existing digital competency frameworks, emphasizing the growing need for educators to develop digital skills for effective AI integration in education. The study proposed specific AI competencies and a revised framework to support teachers in promoting AI education, stressing the significance of continuous professional training for enhancing AI-related technological skills among educators.

### 1.1 | Challenges and implications of artificial intelligence for teacher education

While AI holds great influence in teacher education, it holds challenges related to training, privacy, equity, and ethical concerns. Addressing these challenges and embracing the implications of AI is essential to ensure its positive impact on the education sector (Holmes & Porayska-Pomsta, 2022). Effective integration of AI tools into classrooms is hindered by the lack of necessary training among many teachers, creating a knowledge gap. Moreover, it's crucial to tackle the moral implications tied to the use of AI in education, encompassing issues concerning data confidentiality and prejudice (Vincent-Lancrin & van der Vlies., 2020). AI-driven personalized learning can exacerbate inequality if not implemented thoughtfully. AI disrupts traditional teaching roles, and this raises concerns about job security, demanding that teachers wrestle with the additional challenge of efficiently overseeing AI tools, from selecting appropriate resources to troubleshooting technical issues (Miao & Holmes, 2021). AI can enhance personalized learning, making education more effective. However, it raises concerns about data privacy and bias in algorithm-driven decision-making. Teachers need to adapt and develop AI skills to remain relevant in evolving classrooms. AI may also change traditional teaching roles, impacting job security. Incorporating AI training in teacher education programs is crucial to prepare educators for the landscape steered by AI in education.

### 1.2 | Contribution and novelty

- Investigates the opportunities, challenges, and the impact of AI within the framework of teacher education.
- Examine the consequences of adopting AI technology on teaching practices, specifically focusing on how AI supports classroom management.
- Examine how AI impacts the digital literacy of educators and computer skills, and how AI promotes digital knowledge acquisition for accessing educational resources.
- The novelty of the study is rooted in its exploration of the diverse effects of AI on teacher education, offering insights into both opportunities and challenges and shedding light on the transformative implications for the future of educational practices and pedagogy.

The following sections of the paper are arranged in the following manner: In Section 2, the literature pertinent to the study is reviewed, Section 3 delves into the conceptual framework underlying the study is elucidated. Section 4 furnishes a comprehensive explanation of the research methods, Section 5 showcases the findings, and Section 6 delves into a discussion of the study. The paper concludes in Section 7, encompassing limitations and future scope.

## 2 | LITERATURE REVIEW

In this section, an exhaustive survey of previous research concerning the integration of AI in the realm of learning and instruction is presented. The examination focuses on the impacts of AI on classroom management and teaching support, elucidating AI's function in establishing inclusive and accessible learning environments, and the enhancement of personalized teaching methods through AI, and identifies the research gap.

### 2.1 | AI's impact on classroom management and teaching support

Murphy (2019), discussed how AI can help K-12 teachers and their teaching, not by taking their jobs but by making them better at what

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they do. The study points out three helpful ways AI can be used. They are smart tutoring systems, grading essays automatically, and giving early warnings. Chen et al. (2020) wanted to explore how AI is changing education. They found that AI is being used a lot in education, especially by schools and colleges. It commenced with computers and technologies related to computing, then transitioned to online learning systems. In conclusion, they're even using robots and chatbots to help teachers. These new tools help teachers do administrative tasks like grading faster and better. They also make learning more personalized, so it fits each student's needs. This makes learning more enjoyable and better quality. Kim et al. (2020) performed a study to grasp the viewpoints of students on AI teaching assistants within the sphere of digital learning. They used a digital questionnaire conducted over the internet to ask students about their opinions. The main findings showed that there is a higher likelihood of students accepting AI teaching assistants if they discover them useful and easy to communicate with. These results suggest that using AI teaching assistants in education is a good idea. Seo et al. (2021) intended to investigate the perceptions of students and teachers regarding the influence of AI systems on the interaction between learners and instructors in digital educational settings. Interviews were conducted to gather immediate reactions from 12 learners and 11 educators focusing on various applications of Artificial Intelligence in digital education. Participants saw AI as enabling personalized engagement on a large scale, yet also as a potential for transgressing interpersonal limits. Worries were expressed regarding matters of accountability, autonomy, and monitoring problems. Chaudhry and Kazim (2022) discussed how AIEd focused in terms of lessening the burden on educators, offering tailored education for learners, transforming evaluation methods, and creating smart instructional systems. It also delved into the moral aspects of AIEd and the possible influence of the COVID-19 crisis on the future of AIEd's study and implementation. The study reviewed the latest research in AIEd and discussed the focus of this research. The investigation found that AIEd had the potential to transform education through alleviating the work burden on educators, offering customized educational journeys for learners, transforming evaluation methods, and creating smart teaching systems.

## 2.2 | AI's role in promoting inclusive and accessible learning environments

Vincent-Lancrin and van der Vlies. (2020) wrote a paper to contribute to the G20 AI dialogue. In their study, they discussed how AI was changing education and presented two challenges. First, they investigated using AI to make education better both in the classroom and in the education system. Second, they mentioned the importance of preparing students for jobs in a world with more technology. They explained how AI could make learning more personal and support students with special needs. To make the most of AI, it was important that people trusted it and how it was used. This led to new policy challenges related to "trustworthy AI," which involved protecting data and ensuring it was used fairly and without bias. Chan (2023) conducted a study to create a policy on AI education for advanced education. This was achieved by investigating how text-generative AI technologies are viewed and the implications they entail. The study involved the participation of 457 students, along with 180 teachers and staff members. The review identified diverse the influence of Artificial Intelligence on pedagogical methods and the learning process at the university level. Furthermore, the study recommends that students should be actively involved in shaping and putting this policy into action. Toyokawa et al. (2023) sought to explore the difficulties and potential of implementing AI in the context of inclusive education in Japan. The Learning and Evidence Analysis Framework (LEAF), a learning environment enhanced by learning analytics, was employed in the study. Active Reading was used as a representative learning task in the research, which was conducted in a resource room with two students. The study scrutinized the learning logs in the LEAF system while each student carried out a specific learning task. The research identified distinct learning behaviours from the logs and delved into the challenges and prospective benefits of learning with AI technology, taking into account the role of human intervention in facilitating inclusive educational practices. McArdle (2022) aimed to comprehend the manner in which upcoming pedagogical methods could advocate for fairness and uniformity while simultaneously adopting AI systems oriented toward automation, standardization, and efficiency. The study was rooted in conceptual involvement, identifying significant obstacles for upcoming implementation, and summarizing the most recent investigations and studies. It offered a synopsis of the intricate challenges that emerge from the convergence of AI technologies and the necessity to cultivate inclusive educational practices.

## 2.3 | Enhancing personalized teaching methods through AI

Chen (2023) conducted a study to explore how AI can serve as a virtual trainer for implementing innovative teaching methods tailored to specific training requirements. The research revealed that AI tools can be effectively used in various aspects of training, including managing knowledge, assessing needs, organizing training, and providing feedback on outcomes. Through the utilization of AI-based training methodologies, organizations can undergo a transformation into knowledge-driven entities, adept at meeting individualized learning requirements and improving the comprehensive quality of the educational journey. AI tools usher in a transformation in the various stages of training, such as creating knowledge bases, conducting needs assessments, structuring training programs, and offering result feedback. Renz and Hilbig (2020) discussed EdTech companies and how they use data for teaching and learning. They found that these companies use Learning Analytics with AI for personalized learning. The study also investigated how people think about traditional and modern education. While many like the idea of flexible and personalized

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learning with AI, there are problems like who controls the data, uncertainty, and not fully understanding how data works, which make it hard to use these ideas in practice. Maghsudi et al. (2021) explored personalized learning with the aim of designing an effective knowledge acquisition path that aligned with each learner's strengths while mitigating weaknesses to achieve their desired goals. The study utilized AI methods to accurately capture student characteristics, drawing from past experiences and analysing available big data to uncover learner features and similarities. The study examined the challenges associated with AI-based personalized education and proposed potential solutions. These included addressing the challenges posed by the absence of peer interaction, fostering and sustaining learner motivation, promoting diversity, mitigating biases in data and algorithms, among others.

### 2.4 | Research gap

Previous studies predominantly focus on the execution and the influence of AI in the realm of education from institutional and student perspectives. However, there exists a discernible deficiency in comprehending teacher perceptions, perspectives, and readiness regarding the incorporation of AI into their teaching methodologies. Existing studies depict a broad view of AI applications in education, but they lack specific insights into AI tools and technologies that can be tailored to teacher education. There is a need to identify and examine AI applications specifically designed to enhance teacher training and professional development. Although AI's influence on teacher-student interactions is briefly touched upon, there is a research gap concerning how AI affects the dynamics of the teacher-student relationship. The current collection of scholarly works provides a significant understanding of the influence of AI on diverse aspects of education, however, there is a lack of research specifically exploring how AI can be optimally leveraged within teacher education programs. Thus, this study seeks to scrutinize the prospects, hurdles, and potential outcomes of employing AI in the realm of educator training.

Based on the provided literature review, it is evident that previous research has extensively investigated the effects of AI in the field of education from multiple viewpoints, including classroom management, inclusive learning environments, and personalized teaching methods. However, the existing scholarly works show a significant void concerning teacher perceptions and readiness for integrating AI into their teaching methodologies. While existing studies offer insights into AI applications in education, there is a lack of specific focus on AI tools tailored to teacher education and professional development. To address this gap, the current study aims to scrutinize the prospects, hurdles, and outcomes of employing AI in the realm of educator training. By centering on the viewpoints and preparedness of teachers, the study aims to offer a significant understanding of how AI can be best utilized within teacher education curriculums to augment personalized educational journeys for learners and enhance pedagogical methods. To steer this exploration, the subsequent research queries have been devised:

**RQ1.** What are the specific opportunities presented by AI in teacher education, and how do they impact teaching practices?

**RQ2.** What are the primary difficulties linked with the amalgamation of AI into the instruction of educators, and how do they affect the teaching-learning process?

**RQ3.** What are the implications of AI for educator training in the context of syllabus formulation, teaching methodologies, and learner results?

## 3 | CONCEPTUAL FRAMEWORK

The conceptual framework underpinning this study explores the intricacies of the multifaceted landscape of AI in the domain of teacher education. The framework (Figure 1) seeks to elucidate the opportunities, challenges, and broader implications that AI presents within the realm of teacher education. As the demand for skilled and adaptable educators continues to rise, the fusion of AI and teacher training stands at the forefront of educational innovation. In the following discussion, we will explore the six key hypotheses encapsulated within this conceptual framework, each shedding light on various dimensions of AI's influence on teacher education.

### 3.1 | AI technology and classroom management

AI analyses each student's learning needs and capabilities, empowering teachers to personalize their instruction for individual students. This personalization can help to keep students engaged and motivated, reducing disruptive behaviour in the classroom (Kim et al., 2020). Assessment tools powered by AI can rapidly evaluate student performance on assignments and tests, freeing educators from labour-intensive evaluation duties. This allows educators to

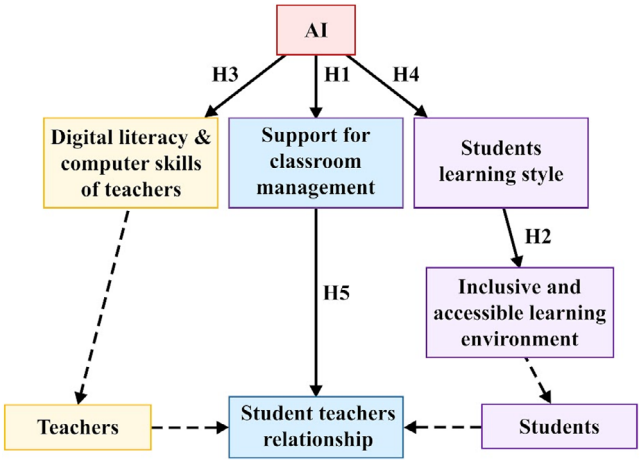


**FIGURE 1** Conceptual framework.

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focus more on instruction and student interaction. AI systems automate attendance tracking and alert teachers to the absence of students. This helps educators manage attendance more efficiently and address any potential issues with truancy. AI assists in optimizing classroom schedules and resource allocation, ensuring that teachers have access to the required materials and facilities when needed, which can improve the overall efficiency of the teaching environment (Chatterjee & Bhattacharjee, 2020). AI analyses student data to identify potential issues or challenges in real-time. For example, it can flag students who may be falling behind or struggling with certain concepts, allowing teachers to intervene and provide additional support before problems escalate (Nazaretsky et al., 2022).

**H1.** Adoption of AI technology positively influences teaching by providing support for classroom management.

## 3.2 | AI technology and learning environment

AI technology can provide educators with valuable perspectives through data analysis, aiding them in making informed decisions concerning classroom management strategies and teaching approaches (Mohammed & Nell'Watson, 2019). AI provides assistive technologies including real-time record services, text-to-speech capacity, and individually tailored learning experiences. These empower students with disabilities or specific learning needs to interact with course content through different formats and at their own pace. AI can create accessibility tools such as AI interfaces, educational chatbots, and assistive robot mentors to support accessible and inclusive education (Miao & Holmes, 2021). AI helps teachers work more efficiently by automating tasks that consume a significant amount of time, such as grading, allowing them to concentrate on providing personalized instruction and support to students. AI provides teachers with a valuable understanding of student learning, empowering them to customize instructional approaches to cater to each learner's requirements and provide real-time feedback (Vincent-Lancrin & van der Vlies., 2020). AI can create an inclusive educational environment to guarantee that all learners, irrespective of impairments or particular educational necessities, have fair access to scholastic materials.

**H2.** AI educates teachers in the process creating a more inclusive as well as accessible learning environment tailored for students.

## 3.3 | AI technology and digital knowledge

Imparting knowledge about AI to students is crucial for fostering digital literacy skills, and analytical thinking abilities, and developing them for future academic endeavours and success in careers. Students are equipped with a fundamental understanding of AI systems, enabling them to engage with and conceptualize ideas about AI methodologies in a secure, accountable, and moral way (Pedro et al., 2019). The study of AI also motivates students to analyse and assess question structures, process complex information, challenge assumptions, and contemplate the ethical considerations linked to the utilization of AI technology. The quantity and quality of communication among learners and educators are positively affected by AI mechanisms (Kateryna et al., 2020). AI proficiency can furnish learners with the necessary understanding and abilities required to pursue vocations in technology, data science, and other in-demand industries. AI also enables teachers to leverage AI to enhance the assistance they provide to individual students when faced with time constraints. AI enhances the adaptability of educational materials to the strengths of students and requirements (Quaicoe & Pata, 2020). AI can positively influence the improvement in digital literacy and computer skills of teachers, as well as promote more digital knowledge to access teaching sources. AI can help teachers and students develop the competencies needed to thrive in an AI-driven learning environment.

**H3.** AI positively influences the improvement in digital literacy and computer skills of teachers as well as promotes more digital knowledge to access teaching sources.

## 3.4 | AI technology and learning styles

AI is thought to positively impact the ability to identify and comprehend individual students' learning styles. Learning styles refer to the various approaches individuals prefer and favour to obtain active information. Common learning styles encompass visual, auditory, kinaesthetic, and reading/writing (Chan, 2023). AI can potentially use data and insights gathered from students' interactions with educational materials and technologies to determine their learning preferences. This information can aid in customizing educational content and strategies to better align with each student's specific needs. AI can offer personalized suggestions, such as which teaching materials to use, how to structure lessons, or when to offer supplementary assistance to particular learners (Chen et al., 2020). Traditionally, teachers have relied on a limited set of instructional techniques. However, AI can furnish educators with insights and recommendations on how to adapt their teaching approaches enabling educators to address the diverse learning styles and demands of their students (Almasri, 2022).

**H4.** AI impacts positively on identifying students' learning styles and helps teachers in adopting different teaching methods.

## 3.5 | AI technology and interactions

AI can assist in customizing educational content based on specific requirements and individual students' learning styles. Through the

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analysis of student data, AI can recommend specific resources, assignments, or study materials, allowing teachers to provide more personalized guidance to each student. This personalization shows students that their unique learning needs are being acknowledged and addressed, enhancing their trust in their teachers (Guilherme, 2019). AI can facilitate better communication between teachers and students through chatbots or virtual assistants. These tools can answer common questions, provide reminders, and offer support, freeing up teachers to focus on more meaningful interactions. Students may find it easier to approach their teachers when they have more pressing concerns, leading to stronger relationships (Shakhnoza, 2023). AI-powered assessment tools can offer students instant feedback on their performance, aiding their comprehension strengths and weaknesses. Teachers can use this feedback to engage in meaningful discussions with students about their progress, setting goals and strategies for improvement. Such interactions create a supportive and constructive learning environment (Ayuwanti & Siswoyo, 2021).

**H5.** AI fosters teacher-student relationships by improving everyday interactions.

The proposed conceptual framework, with its five hypotheses, underscores the dynamic nature of AI's role in teacher education. These hypotheses collectively unveil the intricate interplay between AI and the future of educators, shedding light on the myriad opportunities, challenges, and implications that arise.

## 4 | METHODOLOGY

The methodology section outlines the process used by the research to explore the current analysis problem and objective, in addition to presenting its findings based on the information collected throughout the survey period. This section also demonstrates how the final article findings will be attained in a manner consistent with achieving the paper's goal. This article used quantitative techniques and primary sources to tackle the main investigation goals. Thus, the methods that were employed during the investigation process are discussed in this chapter. The study discusses the research methodology, intended demographic, sample selection, data gathering instruments and process, and data evaluation in this segment.

### 4.1 | Research design

The investigation employed a descriptive quantitative approach to systematically assess the present condition of AI incorporation in educator training. Through quantitative studies, it aimed to gain statistical insights into the application of AI methodologies in educational procedures and their influence on educator-learner interactions and learning outcomes. Research objectives included identifying opportunities, analysing challenges, and evaluating AI's influence on teacher education curricula. The study focused on students and teachers from five academic institutions in China, using a multi-cluster approach and purposive sampling. Data were collected via structured questionnaires from 202 students and 68 staff and analysed using SPSS software.

### 4.2 | Target population

The target population of this study encompasses both students and teachers, recognizing AI integration's transformative impact on education. By including this diverse group, the study aims to comprehensively understand AI's influence on the educational landscape. Through a multi-cluster approach, the research ensures a robust representation of participants, developing a rich pool of experiences and insights. Insights gained from this investigation have the potential to inform educational practices globally, paving the way for effective AI integration in teacher education programs worldwide. Specifically, the study focuses on college students pursuing undergraduate or postgraduate degrees and employees working in educational institutions in China.

### 4.3 | Data collection tool and procedure

This study employed a structured questionnaire as its primary data collection tool. This questionnaire aimed to capture insights from participants regarding their experiences and perceptions regarding the opportunities, difficulties, and consequences of AI in educator training. Employing a five-point Likert scale that spans from 1 for "strongly disagree" to 5 for "strongly agree," participants could express their views effectively. Before the main data collection, a pilot study involving a subset of the sample (10 staff and 10 students) was conducted to refine the questionnaire's validity and reliability based on feedback. During this pilot study, participants were requested to fill out the survey and offer commentary on its lucidity, comprehensibility, and relevance. Their input was carefully analysed to identify any areas of ambiguity or confusion in the questionnaire, allowing for necessary revisions to enhance its effectiveness. Subsequently, the main data collection phase involved distributing the questionnaire using a multi-cluster approach and purposive sampling. A total of 350 questionnaires were distributed, with 250 targeted at students and 100 at staff. The questionnaire consisted of two sections: one gathering demographic information and the other addressing the investigation's objectives. Following filtration, 202 student and 68 teacher questionnaires were deemed suitable for analysis. Table 1 provides a detailed overview of a sample group comprising college students. It presents data on various demographic factors such as gender, age groups, degree levels, and the universities they attend.

Table 2 provides a detailed overview of the staff demographics within the specified organizations. The gender distribution among staff members is equal, with 50.00% being male and 50.00% female.

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**TABLE 1** Sample description of college students.

| Sample description of students | | Number | Percentage |
|---|---|---|---|
| Gender | Male | 100 | 49.50 |
| | Female | 102 | 50.50 |
| Age groups | 18–21 years | 151 | 74.75 |
| | Above 22 years | 51 | 25.25 |
| Degree | Bachelor's degree | 95 | 47.03 |
| | Master's degree | 107 | 52.97 |
| Universities | Peking University | 43 | 21.29 |
| | Southwest Jiaotong university | 37 | 18.32 |
| | Xi'an Jiaotong-Liverpool University | 42 | 20.79 |
| | Northeast Normal University | 39 | 19.31 |
| | Hangzhou Normal university | 41 | 20.30 |

**TABLE 2** Demographic description of staffs.

| Category | Sub-category | Frequency | Percentage |
|---|---|---|---|
| Gender | Male | 34 | 50.00 |
| | Female | 34 | 50.00 |
| Age | Less than 30 years | 5 | 7.35 |
| | 30–45 years | 55 | 80.88 |
| | More than 45 years | 8 | 11.76 |
| Profession | Teachers | 20 | 29.41 |
| | Instructors | 17 | 25.00 |
| | Academic administrators | 17 | 25.00 |
| | Experts/policymakers | 14 | 20.59 |
| Experience | Under 5 years | 5 | 7.35 |
| | 5–15 years | 49 | 72.06 |
| | More than 15 years | 14 | 20.59 |
| Nationality | China | 59 | 86.76 |
| | Others | 9 | 13.24 |

## 4.4 | Representativeness of the sample

By encompassing both students and teachers from various educational institutions in China, the study aims to capture diverse views on the incorporation of AI in educator training. The use of a multi-cluster approach in sampling ensures a robust representation of participants, accounting for potential variations across different academic institutions and roles within the educational sector. This approach helps mitigate biases that may arise from selecting participants from a single institution or demographic group, thereby enhancing the sample's representativeness. Additionally, the pilot study conducted with a subset of the sample allowed for refining the questionnaire's validity and reliability, ensuring that the data collected accurately reflects participants' experiences and perceptions. The feedback obtained during the pilot study helped identify and address any potential shortcomings in the questionnaire, further enhancing the representativeness of the data collected. Overall, the combination of purposive sampling, a multi-cluster approach, and a pilot study contributes to the representativeness of the sample, enabling the study to offer a significant understanding of the prospects, hurdles, and consequences of AI for teacher education in China.

## 4.5 | Data analysis

The collected information experienced analysis through SPSS software. The consistency and dependability of the measuring scales utilized for assessing various components of the research are evaluated using reliability analysis. In this study, five hypotheses were formulated to examine the influence of AI technology on diverse facets of teacher training. Hypothesis 1 posits that the integration of AI technology positively influences teaching by supporting classroom management, linking the variable "Adoption of AI technology" with "Classroom management." Hypothesis 2 suggests that AI education enhances teachers' ability to establish a more comprehensive and reachable educational setting customized for learners, associating "Adoption of AI technology" with "Inclusive and accessible learning environment." Hypothesis 3 explores the effect of AI on improving digital literacy and computer skills among teachers, connecting "Adoption of AI technology" with "Digital literacy and computer skills." Hypothesis 4 investigates how AI impacts the identification of students' learning styles and the adoption of different teaching methods, involving "Adoption of AI technology" and "Students' learning styles." Finally, Hypothesis 5 examines the role of AI in fostering teacher-student relationships by improving everyday interactions, linking the "Adoption of AI technology" with the "Teacher-student relationship." By establishing these associations, the research intends to methodically scrutinize the impact of AI technology on multiple aspects of teacher training. The outcomes of the reliability examination on the various research elements are presented in Table 3.

With Cronbach's alpha values consistently exceeding 0.70, all of the constructs demonstrate good dependability. They are categorized as "Very Reliable" in the analysis, which implies that the measuring scales used to evaluate these constructs are reliable and consistent for the investigation. In assessing the validity of the measurement procedure, the study employed the criterion of construct validity. This

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**TABLE 3** Reliability and validity test.

| Variables | Reliability test | Decision | Validity test | Decision |
|---|---|---|---|---|
| Classroom management | 0.874 | Reliable | 0.777 | Valid |
| Inclusive and accessible learning environment | 0.881 | Reliable | 0.783 | Valid |
| Digital literacy and computer skills | 0.759 | Reliable | 0.712 | Valid |
| Students' learning styles | 0.748 | Reliable | 0.704 | Valid |
| Teacher-student relationship | 0.895 | Reliable | 0.796 | Valid |
| Adoption of AI technology | 0.892 | Reliable | 0.793 | Valid |

**TABLE 4** Descriptive statistics.

| Variables | Mean | Median | Max. | Min. | Std. dev. |
|---|---|---|---|---|---|
| Classroom management | 26.50 | 26.51 | 27.58 | 25.53 | 0.58 |
| Inclusive and accessible learning environment | 25.09 | 25.12 | 26.34 | 23.91 | 0.67 |
| Digital literacy and computer skills | 26.11 | 26.11 | 27.07 | 25.26 | 0.52 |
| Students' learning styles | 25.79 | 25.78 | 26.74 | 24.90 | 0.50 |
| Teacher-student relationship | 24.11 | 24.09 | 25.32 | 22.67 | 0.73 |
| Adoption of AI technology | 20.70 | 20.73 | 21.56 | 19.69 | 0.51 |

criterion ensures that the measurements accurately reflect the underlying theoretical constructs of interest, thereby boosting the trustworthiness and significance of the study's conclusions. The validity test results, with all constructs scoring above 0.70, demonstrate satisfactory construct validity, affirming the accuracy and appropriateness of the measurement procedure. Table 4 exhibits the descriptive statistics for the principal variables in the research, offering valuable insights into their characteristics. Classroom Management exhibits a mean score of 26.50, indicating a central tendency around this value, with minimal variability as suggested by a narrow standard deviation of 0.58. Similarly, Inclusive and Accessible Learning Environment demonstrates a mean of 25.09, with a slightly wider range of values as reflected in its standard deviation of 0.67.

Digital Literacy and Computer Skills, with a mean of 26.11, and Students' Learning Styles, with a mean of 25.79, both show relatively consistent scores across participants, as evidenced by their tight distributions and low standard deviations of 0.52 and 0.50, respectively. In contrast, Teacher-Student Relationship exhibits a slightly broader range of values, as indicated by its mean of 24.11 and a standard deviation of 0.73, suggesting more variability in participants' perceptions of this variable. These descriptive statistics provide a comprehensive overview of the central tendencies, variability, and distribution of scores for each key variable, aiding in the interpretation of the research outcomes.

## 5 | RESULTS

The results section of the paper delves into the findings and outcomes of the investigation into the occasions, challenges, and implications of combining AI into teacher education. This section presents an analysis related to the influences of AI on teacher adaptation for classroom management and AI technology in teaching practice and student achievement.

### 5.1 | Integration of AI technology and classroom management by teachers

AI technology's adoption in learning has significantly impacted teachers' classroom management. AI tools assist educators in many ways, from streamlining administrative tasks like automating grading to ensuring personalized learning resources. This automation reduces teachers' workload, allowing them to focus more on student engagement and instruction. AI also enables data-driven insights, helping teachers identify at-risk students and modify interventions.

Figure 2 illustrates the sample's perception of incorporating AI technology in learning. Most of the respondents concurred that integrating AI technology in education provides vital technological support to teachers (agree 90%, disagree 3%, and neutral 7%), saves time and creates a better lesson plan (agree 80%, disagree 0%, and neutral 20%), foster teacher-student communication and interaction (agree 89%, disagree 1%, and neutral 10%), ensure the classroom offers great learning conditions (agree 89%, disagree 1%, and neutral 10%), creating and supplementing content (agree 75%, disagree 5% and neutral 20%), productivity and efficiency (agree 55%, disagree 40%, and neutral 5%), and personalized learning (agree 60%, disagree 30%, and neutral 10%).

The highest score obtained pertains to the perception that AI provides vital technological support to teachers, with a significant majority (90%) of respondents in agreement. This indicates a widespread acknowledgment of the crucial role AI plays in assisting educators with various teaching tasks. 80% of respondents agree that AI helps save time and create better lesson plans. This indicates that AI is perceived as a beneficial instrument for enhancing efficiency in lesson

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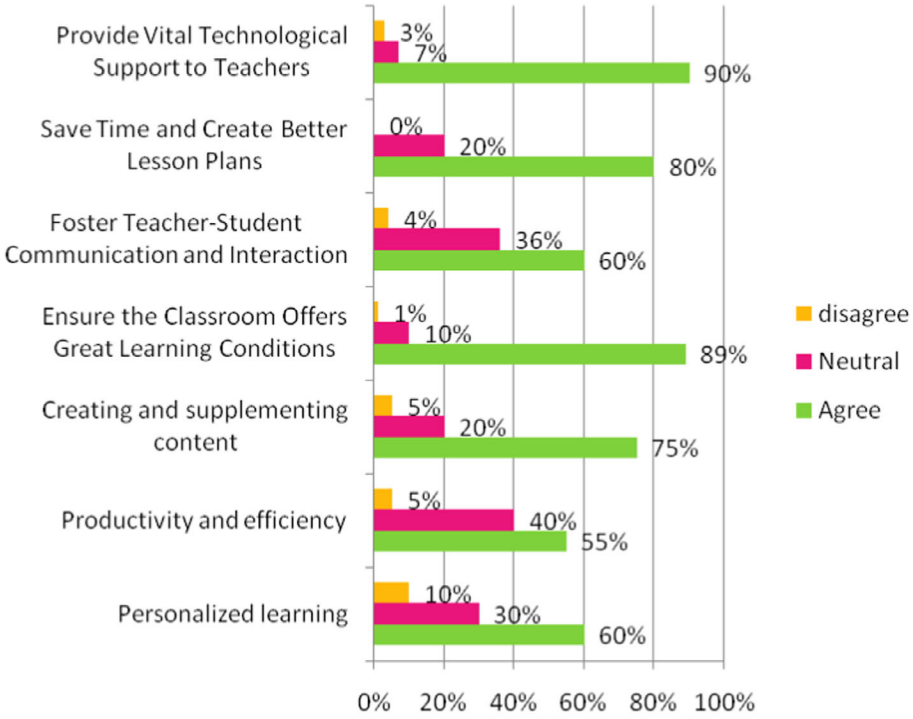


**FIGURE 2** Participants' viewpoints regarding the incorporation of AI technology in education.

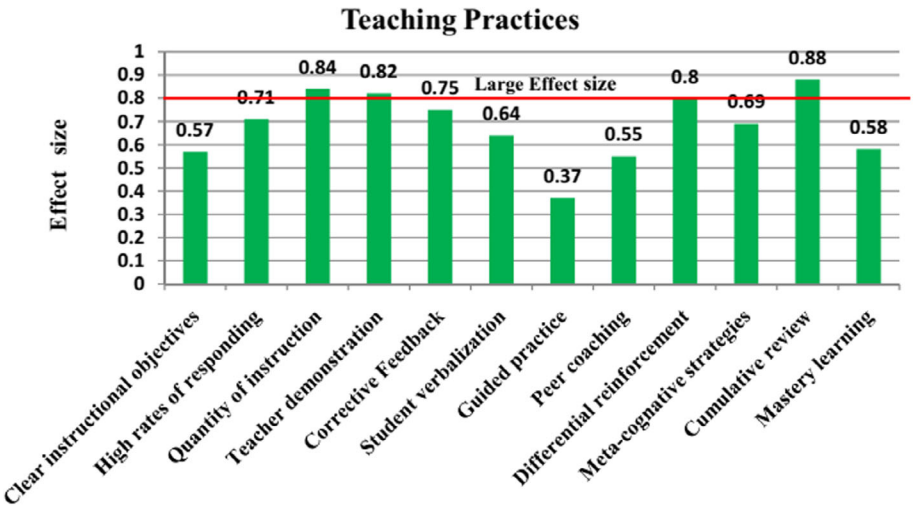


**FIGURE 3** Impact of teaching competencies on students' achievement.

planning. Opinions are more divided on whether AI fosters teacher-student communication and interaction, with 60% agreeing, 36% neutral, and 4% disagreeing. This could suggest that while some see AI as a facilitator of communication, others may have reservations or lack enough information to form an opinion. Most respondents (89%) agree that AI ensures the classroom offers great learning conditions. This shows a positive understanding of the influence of AI on the learning environment. Opinions are also divided on the content creation and supplementation aspect, with 55% agreeing, 30% neutral, and 15% disagreeing. This could indicate varying experiences or perceptions of the effectiveness of AI in content creation and supplementation. Productivity and efficiency aspect has the most balanced view, suggesting that experiences and perceptions of the impact of AI on productivity and efficiency may vary widely among respondents. Similarly, opinions on personalized learning are divided, indicating a need for further exploration of this aspect.

## 5.2 | AI technology in teaching practices and student achievement

AI technology has revolutionized teaching practices, enhancing student achievement. Adaptive learning platforms use AI algorithms to personalize curriculum, ensuring each student's individual needs are met. AI-driven analytics bring forth educators with information according to the performance of students, enabling timely interventions. Virtual tutors powered by AI offer additional support. This technology promotes self-paced learning and engagement.

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Figure 3 delivers the impact of various teaching competencies on students' achievement, with associated numerical values indicating the strength of their influence. Quality of instruction (0.84) stands out as the most influential factor, emphasizing that the overall effectiveness of teaching methods significantly enhances students' achievement. Cumulative review (0.88) and Differential reinforcement (0.80) show strong positive effects on student achievement, indicating that regular review and personalized reinforcement strategies contribute substantially to learning outcomes. Teacher demonstration (0.82) also plays a pivotal role, indicating that effective demonstration of concepts positively influences student achievement. Guided Practice (0.37) has a weaker effect, and Peer Coaching (0.55) has a moderate effect on achievement. Furthermore, qualities of instruction, cumulative review, differential reinforcement, and teacher demonstration have a larger effect on students' achievement.

Cumulative review and quantity of instruction are the most effective teaching practices according to Figure 3. Cumulative Review involves revisiting previously learned material over time, which helps reinforce and consolidate learning, leading to better long-term retention of information. If this were the most effective practice, it would indicate that regular review and reinforcement of past material significantly enhances student achievement. Quality of Instruction refers to how well the teaching is delivered. It includes elements like the teacher's knowledge of the subject, their ability to clearly explain concepts and their use of effective teaching strategies. If this were the most effective practice, it would suggest that how material is taught is crucial for student achievement. Teacher demonstration and differential reinforcement also have high effect sizes, indicating they are quite effective. Teacher demonstration is a direct instructional method where the teacher models the desired behaviour or skill and students learn by observing. This method can have a high effect size as it provides a lucid exemplar for learners to emulate, which can augment their comprehension and proficiency of the skill. Differential reinforcement, on the other hand, is a behaviour management strategy used in Applied Behaviour Analysis. It involves reinforcing desirable behaviours while ignoring or not reinforcing undesirable behaviours. This strategy can be highly effective in promoting positive behaviour change. Both strategies, when used appropriately, can significantly enhance the learning experience and outcomes. Other teaching practices like clear instructional objectives, high rates of response, corrective feedback, student verbalization, guided practice, peer coaching, meta-cognitive strategies, and mastery learning also contribute positively to students' achievement, but their effect sizes are relatively lower. The red line at 0.8 is used as a benchmark to indicate a "large" effect size. Any teaching practice with an effect size above this line could be considered highly effective.

## 5.3 | Hypothesis testing

### 5.3.1 | Goodness of fit

The appropriateness of the study's comprehensive model is assessed using a variety of fit indices and standards, as outlined in Table 5. For the $\chi^2$/df, the value of 3.521 is beneath the criterion of 5, signifying a relatively satisfactory fit. The RMSEA, with a value of 0.052, which is less than or equal to the limit of 0.08, shows a good fit for the model. The NFI value of 0.96 surpasses the limit of 0.9, indicating a good fit. Likewise, the NNFI value of 0.98 exceeds the limit of 0.9, indicating a good fit.

**TABLE 5** Evaluation of overall model fit indices and criteria.

| Fit index | Value | Good fit criterion |
|---|---|---|
| $\chi^2$/df (chi-square normalized by degrees of freedom) | 3.521 | Less than 5 |
| RMSEA (Root mean square error of approximation) | 0.052 | ≤0.08 |
| NFI (Normed fit index) | 0.96 | ≥0.9 |
| NNFI (Non-normed fit index) | 0.98 | ≥0.9 |
| CFI (Comparative fit index) | 0.94 | ≥0.9 |
| IFI (Incremental fit index) | 0.95 | ≥0.9 |
| RFI (Relative fit index) | 0.92 | ≥0.9 |
| RMR (Root mean square residual) | 0.032 | ≤0.05 |
| Standardized RMR | 0.034 | ≤0.05 |
| GFI (Goodness of fit index) | 0.92 | ≥0.9 |
| Overall model fit (LISREL estimates) | Met a goodness of fit standard | Research models expected to be robust |

The CFI value of 0.94 satisfies the standard of 0.9, signifying a good fit. With a value of 0.95, the IFI surpasses the limit of 0.9, indicating a good fit. The RFI value of 0.92 fulfils the standard of 0.9, signifying a good fit. The RMR value of 0.032 is beneath the limit of 0.05, indicating a good fit. With a value of 0.034, the standardized RMR satisfies the standard of 0.05, indicating a good fit. The GFI value of 0.92 fulfils the standard of 0.9, indicating a good fit. In summary, the model meets the goodness-of-fit standards, suggesting that the research models are anticipated to be sturdy and provide dependable representations of the data.

### 5.3.2 | Measurement model

The study's measurement model evaluates the discriminant validity using the Fornell & Larcker criterion, which contrasts the square root of the average variance extracted (AVE) of each construct with the correlations among constructs. Discriminant validity guarantees that each construct is unique from others and quantifies a distinct facet of the phenomenon being studied. In the Fornell & Larcker criterion, the square root of the AVE for each construct should surpass the correlation between that construct and other constructs. Observing Table 6, the diagonal elements depict the square root of the AVE for each

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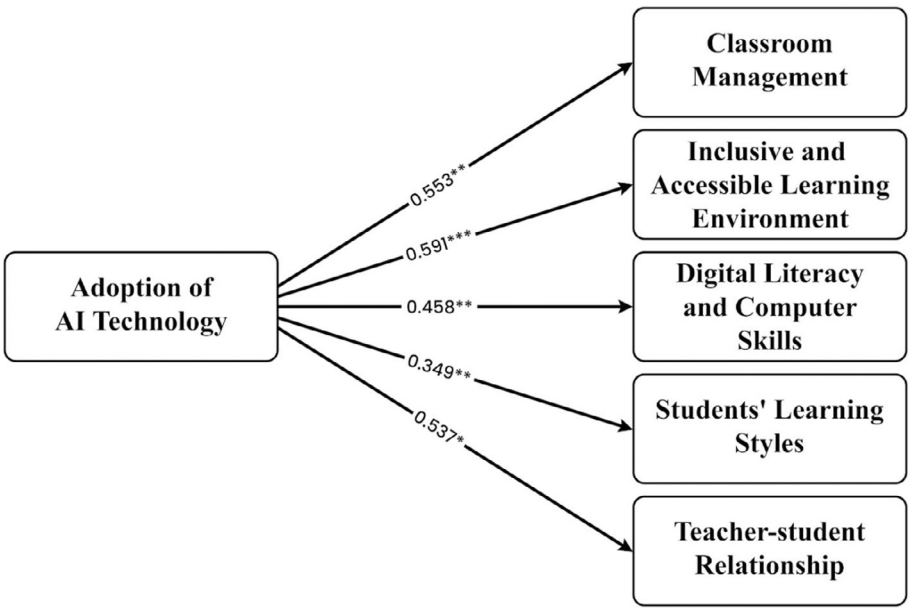
**TABLE 6** Discriminant validity by Fornell & Larcker.

| Variables | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Classroom management (1) | 0.762 | | | | | |
| Inclusive and accessible learning environment (2) | 0.563 | 0.761 | | | | |
| Digital literacy and computer skills (3) | 0.548 | 0.498 | 0.830 | | | |
| Students' learning styles (4) | 0.210 | 0.528 | 0.573 | 0.888 | | |
| Teacher-student relationship (5) | 0.530 | 0.602 | 0.514 | 0.170 | 0.801 | |
| Adoption of AI technology (6) | 0.664 | 0.609 | 0.697 | 0.678 | 0.672 | 0.802 |

**FIGURE 4** Path analysis results.




construct, while the off-diagonal elements illustrate the correlations among constructs.

The square root of the AVE for each construct is contrasted with the correlations between that construct and other constructs. As the diagonal values are consistently greater than the off-diagonal values, it verifies the existence of discriminant validity. This signifies that each construct quantifies a unique facet of the phenomenon being studied, without substantial overlap with other constructs, fortifying the overall validity of the measurement model.

### 5.3.3 | Path analysis and hypothesis testing

In the path analysis results of this study, each hypothesis corresponds to a specific path in the path analysis model, representing the proposed relationship between the adoption of AI technology and various educational outcomes. Figure 4 displays the standardized path coefficients for each path, indicating the intensity and orientation of the association between the predictor variable (adoption of AI technology) and the outcome variable. The path analysis results figure provides an overall summary of the hypothesized relationships between the adoption of AI technology and various educational outcomes, along with their statistical significance and strength of association.

Table 7 outlines the outcomes of the hypothesis examination for the suggested associations between the adoption of AI technology and various aspects of teaching and the educational environment. Each hypothesis corresponds to a specific path in the path analysis model, representing the proposed relationship between the adoption of AI technology and various educational outcomes. The path indicates the orientation of the association, with "Adoption of AI technology" as the predictor variable and the educational outcome as the outcome variable. $\beta$ represents the standardized path coefficients, indicating the intensity and course of the connection between the predictor and outcome variables. S.E. provides the standard error for each path coefficient, indicating the precision of the estimate. The p-value indicates the statistical significance of each path coefficient, determining whether the connection between the predictor and result variables is significant. To determine the statistical significance of the relationships between variables, t-tests were conducted. The results of these tests as given in Table 7 provide evidence supporting the hypotheses formulated in this investigation. The *t*-value represents the t-statistic associated with each path coefficient, which is used to test the significance of the coefficients.

The examination uncovered statistically significant associations between the adoption of AI technology and all educational outcomes (H1–H5). Specifically, the adoption of AI technology significantly

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**TABLE 7** Hypotheses testing results.

| Hypothesis | Path | $\beta$ | S.E | *p*-value | *t*-value |
|---|---|---|---|---|---|
| H1 | Adoption of AI technology → Classroom management | 0.553 | 0.06 | ** | 8.777 |
| H2 | Adoption of AI technology → Inclusive and accessible learning environment | 0.591 | 0.071 | *** | 7.812 |
| H3 | Adoption of AI technology → Digital literacy and computer skills | 0.458 | 0.07 | ** | 9.837 |
| H4 | Adoption of AI technology → Students' learning styles | 0.349 | 0.06 | ** | 6.735 |
| H5 | Adoption of AI technology → Teacher-student relationship | 0.537 | 0.072 | * | 8.951 |

***$p < 0.001$; **$p < 0.01$; *$p < 0.05$.

influenced classroom management ($\beta = 0.553$, $p < 0.01$), created an inclusive and accessible learning environment ($\beta = 0.591$, $p < 0.001$), positively impacted the improvement of digital literacy and computer skills ($\beta = 0.458$, $p < 0.01$), exhibited a substantial positive impact on identifying students' learning styles ($\beta = 0.349$, $p < 0.01$), and significantly contributed to fostering teacher-student relationships ($\beta = 0.537$, $p < 0.05$). The relatively high *t*-values for all paths indicated the strength of the associations between the predictor variable and the result variables. In summary, the findings support all proposed hypotheses, suggesting that the adoption of AI technology positively influences various aspects of teaching and educational environments, including classroom management, learning environment inclusivity, digital literacy, student learning styles, and teacher-student relationships.

## 6 | DISCUSSION

The incorporation of AI in teacher training has emerged as a transformative force, impacting various dimensions of teaching practices and educational outcomes. This discussion elucidates the findings regarding the examination's role in tackling the research questions and synthesizes them with the insights gleaned from the reviewed literature, providing a coherent narrative of the implications and contributions of the integration of AI in the field of educator training. The study's findings highlight the significant opportunities presented by AI in teacher education and their influence on teaching practices, aligning with RQ1. Across various dimensions, AI technologies offer transformative benefits to educators. Firstly, AI facilitates classroom management by automating administrative tasks such as grading and lesson planning, effectively reducing teachers' workload (Chen et al., 2020; Murphy, 2019). This is demonstrated by the significant affirmative $\beta$ coefficient (0.553, $p < 0.01$) observed in H1, affirming AI's substantial influence on classroom management. Furthermore, AI-driven analytics empower educators with data-driven insights, enabling personalized interventions to support student learning and enhance outcomes (Seo et al., 2021). These insights are further substantiated by the significant positive effects observed in H4 and H5, indicating AI's positive impact on identifying students' learning styles and fostering teacher-student relationships, respectively. Thus, the study elucidates how AI presents specific opportunities in teacher education, shaping and improving teaching practices across various domains, in response to RQ1. Addressing RQ2, the study sheds light on the primary obstacles related to the incorporation of Artificial Intelligence into educator training and their influence on the instruction-learning procedure. Despite the promising opportunities AI presents, several hurdles hinder its seamless integration. Privacy concerns, ethical considerations, and the necessity for adequate training and support for educators emerge as significant challenges (Chan, 2023; Vincent-Lancrin & van der Vlies., 2020). However, the results of the research suggest that the benefits of AI adoption outweigh these challenges. The statistically significant relationships observed between AI adoption and various educational outcomes (H1–H5) highlight the revolutionary capacity of AI in the realm of teacher training, emphasizing its capacity to positively influence teaching practices and student learning experiences despite the challenges it poses. Addressing RQ3, the research probes into the implications of AI for teacher education concerning curriculum design, pedagogical strategies, and student outcomes. The findings highlight the transformative impact of AI adoption across these dimensions. AI-driven adaptive learning platforms personalize learning experiences, ensuring that individual student needs are effectively addressed. This personalized approach enhances student engagement and achievement, as evidenced by the significant positive effects observed in H2 and H3, indicating the creation of inclusive learning environments and improvements in digital literacy and computer skills, respectively. Moreover, AI technologies enable the implementation of innovative teaching methods tailored to specific training requirements, thereby fostering a transformative shift toward knowledge-driven entities (Chen, 2023; Renz & Hilbig, 2020). These findings underscore the importance of integrating AI in teacher education to enhance teaching practices and ultimately improve student outcomes.

*Impact on teaching support and classroom management (H1)*: The outcomes of the investigation align with prior studies by Murphy (2019) and Chen et al. (2020), affirming that AI positively influences teaching support and classroom management. Through the implementation of smart tutoring systems and automation of administrative tasks, AI assists educators in enhancing efficiency and effectiveness in classroom management.

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*Role in creating inclusive and accessible learning environments (H2)*: Consistent with Vincent-Lancrin and van der Vlies. (2020) and Chan (2023), the study supports the notion that AI plays a pivotal role in creating inclusive and accessible learning environments. By providing personalized learning experiences and leveraging text-generative AI technologies, educators can cater to diverse student needs, promoting inclusivity and equity in education.

*Contribution to improving teachers' digital literacy and computer skills (H3)*: The findings align with the research conducted by Kim et al. (2020) and Seo et al. (2021), emphasizing the beneficial influence of AI on improving teachers' digital literacy and computer skills. AI-driven adaptive learning platforms and AI teaching assistants contribute to enhancing teachers' technological proficiency, facilitating effective integration of digital resources in teaching practices.

*Positive influence on identifying students' learning styles and facilitating adoption of diverse teaching methods (H4)*: The study's findings resonate with the research conducted by Chaudhry and Kazim (2022) and McArdle (2022), emphasizing the function of AI in discerning pupils' learning modalities and aiding the assimilation of varied pedagogical techniques. By employing AI tools, instructors can acquire crucial understanding into personal learning predilections, empowering them to customize teaching strategies suitably.

*Role in strengthening teacher-student relationships through improved interactions (H5)*: Furthermore, consistent with the research by Chen (2023) and Renz and Hilbig (2020), the study supports the idea that AI contributes to strengthening teacher-student relationships through improved interactions. By leveraging AI-based training methodologies and personalized learning algorithms, educators can create tailored learning experiences that foster meaningful engagement and communication between teachers and students.

In summary, this study contributes to existing knowledge by providing empirical evidence of the positive impact of AI adoption in teacher education across various dimensions. By aligning with previous research, the results emphasize the revolutionary capability of AI in enhancing teaching practices, promoting inclusivity, and improving student outcomes. These insights have significant implications for theory and practice, highlighting the significance of incorporating Artificial Intelligence tools in teacher training to cater to the changing requirements of students and teachers in the era of digitization.

### 6.1 | Practical implication

The incorporation of AI in teacher education presents a spectrum of possibilities and obstacles, reshaping the landscape of the field. One of the significant opportunities lies in personalized learning experiences for aspiring teachers. AI can analyse individual learning patterns and provide tailored content, enhancing their understanding of teaching methodologies and subject matter. Additionally, AI-powered tools can assist educators in developing innovative teaching strategies, improving classroom management, and assessing student performance effectively. However, challenges also arise, such as the need for educators to acquire digital literacy skills and adapt to new teaching methods. Ensuring equitable access to AI technologies and addressing concerns associated with ensuring privacy and security is vital and presents significant challenges that must be addressed with careful consideration. Moreover, there exists a necessity for continuous teacher training to help educators stay updated with AI advancements and effectively integrate them into their educational enactment. Furthermore, collaborations between scholastic institutions and technology companies can support the broadening of user-friendly AI applications customized to the specific needs of teachers.

### 6.2 | Theoretical implication

AI-driven personalized learning experiences resonate with the constructivist approach, enabling students to construct knowledge through interactive and adaptive engagement with AI tools. This connection fosters a deeper understanding of educational concepts and pedagogical practices, aligning with the principles of constructivism. Moreover, the study has implications for socio-cultural thesis highlights the part of social interplay and cultural surroundings in education. AI can facilitate collaborative learning environments, connecting teachers-in-training with diverse perspectives and expertise globally. This interaction enriches their learning experiences, aligning with the socio-cultural theory's emphasis on social interactions as fundamental to knowledge construction. From a cognitive psychology perspective, integrating AI tools can enhance memory retention and cognitive processing. AI-driven accommodative studying platforms can recognize individual educating styles and preferences, providing tailored content that optimizes cognitive engagement and promotes effective learning.

## 7 | CONCLUSION

The integration of AI in teacher education holds immense promise for transforming teaching and learning practices. By leveraging AI technologies, educators can create more engaging and effective learning environments, enhance student outcomes, and prepare students for success in the digital age. However, it is essential to address potential challenges such as privacy concerns, ethical considerations, and the need for ongoing professional development to ensure the responsible and effective use of AI in education. This study delved into the opportunities, challenges, and implications of AI for teacher education, employing a descriptive quantitative approach to assess the current landscape. Through the analysis, several key findings emerged, aligning with the established hypotheses. The findings underscore the positive influence of AI technology on teaching, including support for classroom management, the creation of inclusive learning environments, and the enhancement of digital literacy among educators. Additionally, AI has proven effective in identifying students' learning styles and fostering stronger teacher-student relationships. However, challenges such as the need for adequate training and addressing ethical considerations remain. Moving forward, the study emphasizes the importance of leveraging AI to enhance teaching practices while also addressing associated challenges to ensure its effective implementation in teacher education curricula. By doing so, harnessing the full

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potential of AI can improve educational outcomes and enrich the teaching-learning experience for both educators and students.

## 7.1 | Limitation

The size of the study's sample might not comprehensively appear for the diverse range of perspectives and experiences within teacher education. The data were collected through structured questionnaires, which are subject to self-report bias. Participants may have provided socially desirable responses or misunderstood some questions, certainly influencing the precision of the findings. Longitudinal research would supply a greater thorough comprehension of the long-term crash of AI in teacher education. The rapid evolution of AI and EdTech may render the findings less relevant or applicable as technology continues to advance. The study's design and interpretation of findings may be influenced by the researchers' perspectives and biases.

## 7.2 | Future scope

Conduct long-term studies to examine the long-term impact of AI on teacher education comfort. Examine how AI-enhanced teaching practices evolve and how they affect the career trajectories of educators. Compare the impact of AI in teacher education across different countries, cultures, and educational systems. Conduct in-depth case studies at the institutional level to explore successful AI implementations in teacher education. Delve into data privacy, equity, and fairness issues associated with AI implementation, and propose policy frameworks to ensure responsible AI use in teacher schooling. Investigation on the amalgamation of AI in teacher professional development programs. Examine how achieving a balance between human and AI instruction can yield optimal results.

### AUTHOR CONTRIBUTIONS

**Zhuo Zhang:** Methodology. **Jia Zhang:** Methodology.

### CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

### DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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